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MULTI-CLASS WASTE OBJECT DETECTION USING YOLO11N: A DEEP LEARNING APPROACH FOR INTELLIGENT WASTE SEGREGATION

Harshpratap Singh Rathore*1, Rishi Atulkumar Modh*2, Kanheyalal Suthar*3, Shubham Singh*4, Sharma Varun Sanjay*5

*1,2,3,4PG Student – Shrimad Rajchandra Institute Of Management & Computer Science, Uka Tarsadia University, Bardoli, Gujarat-394350, India.

*5UG Student – Chotubhai Gopalbhai Patel Institute Of Information Technology, Uka Tarsadia University, Bardoli, Gujarat-394350, India.

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ABSTRACT

Waste detection plays a crucial role in modern waste management systems, especially as the amount of waste generated in urban environments continues to rise. Accurate detection is essential for various real-world applications such as automated waste segregation, recycling assistance, smart waste management, and environmental monitoring. With the advancement of artificial intelligence, deep learning methods have proven to be highly effective in addressing these challenges. In this study, we implemented a deep learning-based object detection method for identifying multiple types of waste objects within images using the TACO (Trash Annotations in Context) dataset, containing 1,500 images, 4,784 annotated objects, and 60 different waste categories. The original COCO-format annotations were converted into the YOLO format, and the dataset was divided into training, validation, and testing subsets. To perform multi-class waste object detection, we trained a YOLO11n model using transfer learning and evaluated its performance using Precision, Recall, mAP@50, and mAP@50-95. The results indicated that the model achieved a Precision of 45.63%, Recall of 11.83%, mAP@50 of 13.45%, and mAP@50-95 of 11.44% on the test dataset. These results demonstrate the feasibility of using YOLO-based deep learning for multi-class waste object detection while also highlighting the challenges caused by class imbalance and limited samples in several waste categories.

Keywords: Waste Object Detection, Deep Learning, YOLO11, Computer Vision, Waste Segregation, TACO Dataset, Object Detection, Transfer Learning.

I. INTRODUCTION

In recent years, Deep Learning (DL) has become one of the most significant advancements in artificial intelligence, leading to innovation in many areas like healthcare, agriculture, education, finance, and environmental management. In the field of waste management, the need for smart and automated systems has increased the use of DL-based methods for various tasks such as waste classification, object detection, automated segregation, recycling assistance, and environmental monitoring. Among these, waste object detection has become especially important because it forms the basis for several intelligent waste management applications. Having accurate waste detection is essential for tasks like automated waste segregation, identifying recyclable materials, supporting recycling processes, monitoring waste collection areas, and improving overall waste management efficiency. Older methods that rely on manual identification or manually created features have not been effective, as they are usually slow, make mistakes, and struggle with real-world problems like overlapping objects, different object sizes, complex backgrounds, poor lighting, and large variations in waste appearance. Deep Learning, especially Convolutional Neural Networks (CNNs), is tackling these issues by automatically extracting meaningful features from images, leading to better detection results in different conditions and making it possible to use these systems on a larger scale.

Even though DL shows great promise for waste object detection, how well it works depends a lot on the underlying detection architecture chosen. Many deep learning models have been used for waste detection tasks, each with its own advantages and drawbacks. YOLO-based architectures have become widely used for object detection because they can identify multiple objects within a single image while simultaneously predicting their

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locations and classes. YOLO11 is designed to provide an effective balance between detection accuracy, computational efficiency, and inference speed. Its ability to process an image in a single detection pipeline makes it suitable for applications where multiple waste objects need to be detected simultaneously. In this study, YOLO11n was selected as the detection model and fine-tuned using transfer learning on the TACO (Trash Annotations in Context) dataset, which contains diverse real-world waste images covering multiple waste categories. The model is evaluated using object detection metrics such as Precision, Recall, mAP@50, and mAP@50-95 to determine its effectiveness in identifying different types of waste objects. Together, these aspects provide a practical deep learning approach for evaluating multi-class waste detection and establish a baseline for future improvements in intelligent waste segregation systems.

The rest of this paper is structured as follows. Section 2 provides an in-depth review of existing literature on the use of deep learning in waste classification and object detection. Section 3 describes the proposed deep learning framework, including the YOLO11 architecture, TACO dataset, and data preparation process. Section 4 explains the experimental setup, dataset division, training configuration, and implementation details, followed by a thorough presentation of the results and evaluation metrics. Finally, Section 5 summarizes the key findings, highlights the performance of the YOLO11-based waste detection approach, and discusses possible future improvements for intelligent waste management and automated waste segregation systems.

II. LITERATURE REVIEW

Waste object detection has been widely studied using computer vision and deep learning methods. Early approaches mainly relied on image classification and handcrafted visual features, but these methods were limited when multiple waste objects appeared within a single image or when objects were partially occluded and surrounded by complex backgrounds. With the rise of deep learning, Convolutional Neural Networks (CNNs) and object detection architectures have become increasingly important for automated waste identification, providing improved capability for locating and classifying objects simultaneously. Researchers have explored dedicated waste datasets, two-stage detectors, single-stage YOLO-based architectures, and specialized approaches for cluttered or low-data environments. This section reviews prior studies, highlighting the datasets, techniques, and models that form the basis for our YOLO-based waste detection study.

Proença and Simões [1] introduced the **Trash Annotations in Context (TACO)** dataset as an open dataset for litter detection and segmentation in real-world environments. The dataset contains images of litter collected under diverse conditions and provides manually annotated objects using a hierarchical taxonomy. In their study, they reported results using Mask R-CNN for instance segmentation and highlighted the difficulty of detecting litter in uncontrolled environments. The authors also emphasized that the relatively small number of annotated images makes waste detection a challenging problem, establishing TACO as an important benchmark for future research in real-world litter detection.

A comparative study on litter detection [2] investigated several state-of-the-art deep learning object detection approaches, including Faster R-CNN, Mask R-CNN, EfficientDet, RetinaNet, and YOLOv5. The study evaluated these approaches on litter datasets and focused not only on detection performance but also on practical considerations such as processing time and memory consumption, particularly for applications on resource-constrained devices. Their findings indicated that YOLO-based detectors were promising for litter detection because of their favorable balance between detection performance and computational efficiency.

Bashkirova et al. [3] introduced the **ZeroWaste** dataset to address the challenges of waste detection in highly cluttered industrial environments. Unlike simplified object-centric datasets, ZeroWaste contains images captured in actual material recovery facilities, where waste objects can overlap, deform, and appear in difficult configurations. Their work demonstrated that real recycling environments create significant challenges for both object detection and segmentation and established cluttered waste sorting as an important research problem for computer vision.

Hong et al. [4] proposed **TrashCan 1.0**, a dataset designed for visual detection of marine debris. The dataset provides both bounding-box and segmentation annotations and was developed to support robust detection of underwater waste. The authors evaluated baseline object detection using Faster R-CNN and instance

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segmentation using Mask R-CNN, demonstrating the importance of specialized datasets for detecting waste under challenging environmental conditions.

Zhou et al. [5] focused on the problem of **limited annotated samples** in automatic waste detection. They proposed a few-shot waste detection framework containing a waste proposal module and a waste classification module with attention mechanisms to improve detection when only a small number of examples are available. Using TACO for evaluation, their framework achieved a mean average precision of 31.16% across 12 waste categories when only 30 instances per category were provided. Their work highlights the major influence of annotation quantity and class scarcity on waste detection performance.

Demetriou et al. [6] performed a broad comparison of single-stage and two-stage detectors for construction and demolition waste detection. The study evaluated 18 detection models using a dedicated dataset containing brick, concrete, and tile objects and also considered normally stacked and heavily adhered waste conditions. Their experiments showed that YOLOv7 achieved strong detection accuracy together with fast inference, while object stacking and adherence could significantly affect detector performance. This study demonstrates the practical value of YOLO-based architectures for automated waste sorting under challenging physical conditions.

Similarly, an improved YOLOv5-based construction waste detection approach [7] incorporated **CBAM, simplified SPPF, and multi-scale detection** to enhance feature extraction and object localization. The improved model was compared with conventional detectors including Faster R-CNN, YOLOv3, YOLOv4, and YOLOv7 and showed improved detection performance on the authors' construction-waste dataset. This work demonstrates how modifying a YOLO architecture can further improve its suitability for specialized waste detection tasks.

Recent work has also explored YOLO-based waste identification for practical waste-management applications. **WasteInNet** [8] proposed a YOLOv7-based deep learning model for identifying different types of waste, emphasizing automated waste-type identification for waste management applications. Such studies demonstrate the increasing use of one-stage YOLO detectors for practical waste recognition systems.

From these studies, it is evident that deep learning-based object detection has become an important approach for automated waste identification and sorting. Two-stage methods such as Faster R-CNN and Mask R-CNN provide strong detection and segmentation capabilities, while YOLO-family architectures offer an attractive combination of localization ability, computational efficiency, and fast inference. At the same time, existing literature shows that waste detection remains challenging because of cluttered scenes, object overlap, visual variation, and severe differences in the number of available samples across waste categories. These findings establish **dataset diversity, class balance, detection architecture, and computational efficiency** as important factors for reliable multi-class waste detection. In this study, we therefore investigate **YOLO11n** on the TACO dataset and evaluate its ability to detect multiple waste objects across **60 annotated waste categories**, providing a baseline for future improvements in intelligent waste segregation systems.

III. METHODOLOGY

This section outlines the methodology adopted in this study to address the problem of multi-class waste object detection in real-world images. We employed a deep learning-based object detection model, specifically the YOLO11n architecture, to identify and localize different types of waste objects within a single image. The model was fine-tuned using the TACO (Trash Annotations in Context) dataset, comprising diverse waste images collected from real-world environments and annotated across multiple waste categories. The dataset provides considerable variation in object type, size, position, background, and scene composition, making the detection task more realistic. A detailed description of the dataset is provided in Section 3.2, while this section first presents the detection model and the overall experimental methodology.

3.1 MODEL

To perform multi-class waste object detection, we selected the YOLO11n architecture as the primary deep learning model. YOLO11n represents a lightweight and efficient object detection architecture designed to detect multiple objects and their corresponding classes within a single image. Its detection framework provides a

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balance between object localization, classification performance, computational efficiency, and inference speed, making it suitable for practical waste detection applications.

3.1.1 YOLO11n

YOLO11n is a lightweight variant of the YOLO11 object detection architecture designed for efficient and fast visual recognition. Unlike conventional image classification models that assign a single class to an entire image, YOLO11n performs object detection by simultaneously predicting the location, class, and confidence of multiple objects present within an image. The architecture consists of convolutional feature extraction layers followed by multi-scale feature processing and a detection head, allowing the model to identify objects of different sizes. In this study, the pretrained YOLO11n model was fine-tuned on the TACO dataset containing 60 waste categories. The model was selected as a baseline for evaluating the feasibility of multi-class waste detection under diverse and cluttered visual conditions.

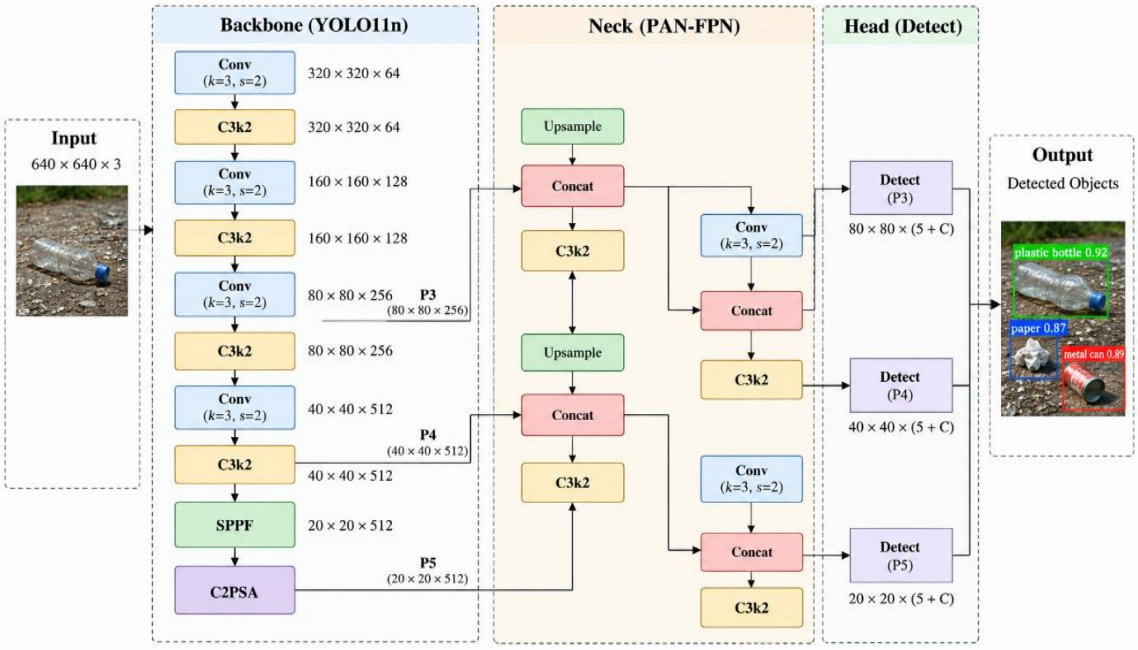


Figure 1: YOLO11n Architecture

3.2 DATASET

To ensure realistic and challenging conditions for waste object detection, we selected the TACO (Trash Annotations in Context) dataset as the primary dataset for this study. TACO is a real-world litter dataset containing images collected from different environments and annotated with waste objects using a detailed category structure. The dataset used in our implementation contains **1,500 images, 4,784 annotated objects, and 60 waste categories**. These categories cover a wide range of waste materials, including bottles, cans, paper products, cartons, plastic objects, bags, containers, metal objects, and other forms of litter.

The original dataset annotations were provided in the **COCO annotation format**. To make the dataset compatible with the YOLO11n training framework, the bounding-box annotations were converted from COCO format into normalized YOLO format. Each image was associated with its corresponding object annotations, including the class identifier and bounding-box coordinates.

Table 1: Dataset Summary of the TACO Dataset

Dataset Name	Total Images	Annotated Objects	Classes	Description
TACO	1,500	4,784	60	Real-world litter and waste images with object annotations

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3.2.1 Dataset Distribution

For our implementation, the TACO dataset was divided into three subsets for model development and evaluation. The complete dataset contains **1,500 RGB images** with **4,784 annotated waste objects** distributed across **60 different categories**. The dataset includes waste objects captured in diverse real-world scenes, with differences in object size, orientation, background, illumination, and degree of object overlap.

The dataset was divided using a fixed random seed into **70% training, 20% validation, and 10% testing data**, resulting in **1,050 training images, 300 validation images, and 150 testing images**. The training subset was used for model learning, the validation subset was used to monitor model performance during training, and the test subset was reserved for final evaluation.

Table 2: Distribution of Dataset Images

Dataset Split	Images	Percentage
Training	1,050	70%
Validation	300	20%
Testing	150	10%
Total	**1,500**	**100%**

The dataset contains substantial variation in the number of annotations among the 60 waste categories. Certain categories contain a large number of annotated objects, whereas several categories contain only a very small number of samples. For example, **Cigarette** contains 667 annotations, **Unlabeled litter** contains 517, and **Plastic film** contains 451 annotations, while **Battery** contains only 2 annotations, **Carded blister pack** contains 1 annotation, and **Plastified paper bag** contains no annotated objects in the downloaded dataset. This uneven distribution introduces a significant class-imbalance challenge for multi-class waste detection.

For our implementation, the converted YOLO dataset was organized into separate image and label directories for training, validation, and testing. The final directory structure contained the corresponding image files, YOLO annotation files, and a data.yaml configuration file defining the dataset paths and all 60 waste categories. The resulting dataset was then used for training and evaluating the YOLO11n object detection model.

IV. EXPERIMENT SETUP

The experiments were conducted using the Ultralytics YOLO framework on Google Colab with NVIDIA Tesla T4 GPU support. The implementation was performed using Python 3.13.15 and PyTorch with CUDA acceleration. The TACO dataset was divided into 70% training, 20% validation, and 10% testing subsets, containing 1,050, 300, and 150 images, respectively. The input images were resized to 640 × 640 pixels during training. A pretrained YOLO11n model was fine-tuned for 30 epochs using a batch size of 16. The model training utilized the AdamW optimizer selected automatically by the training framework. The trained model was evaluated using Precision, Recall, mAP@50, and mAP@50-95 to measure its object detection performance.

V. IMPLEMENTATION & RESULT DISCUSSION

In this section, we present the implementation details of our work. The TACO dataset was used for multi-class waste object detection, covering 60 different waste categories and multiple objects within individual images. The original COCO-format annotations were converted into YOLO-compatible bounding-box annotations before training. To detect and classify waste objects, we utilized the YOLO11n architecture pretrained on a general object-detection dataset and fine-tuned it on the TACO dataset. The model was trained, validated, and finally evaluated on the held-out test dataset to analyze its effectiveness in detecting different types of waste objects. The following subsections describe the model implementation and its detection performance.

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1. YOLO11n

The YOLO11n model was implemented as a lightweight object detection architecture for identifying and localizing multiple waste objects within a single image. During training, the model learned both object localization and class-specific features from the annotated TACO dataset. The model was configured to detect all 60 waste categories defined in the dataset.

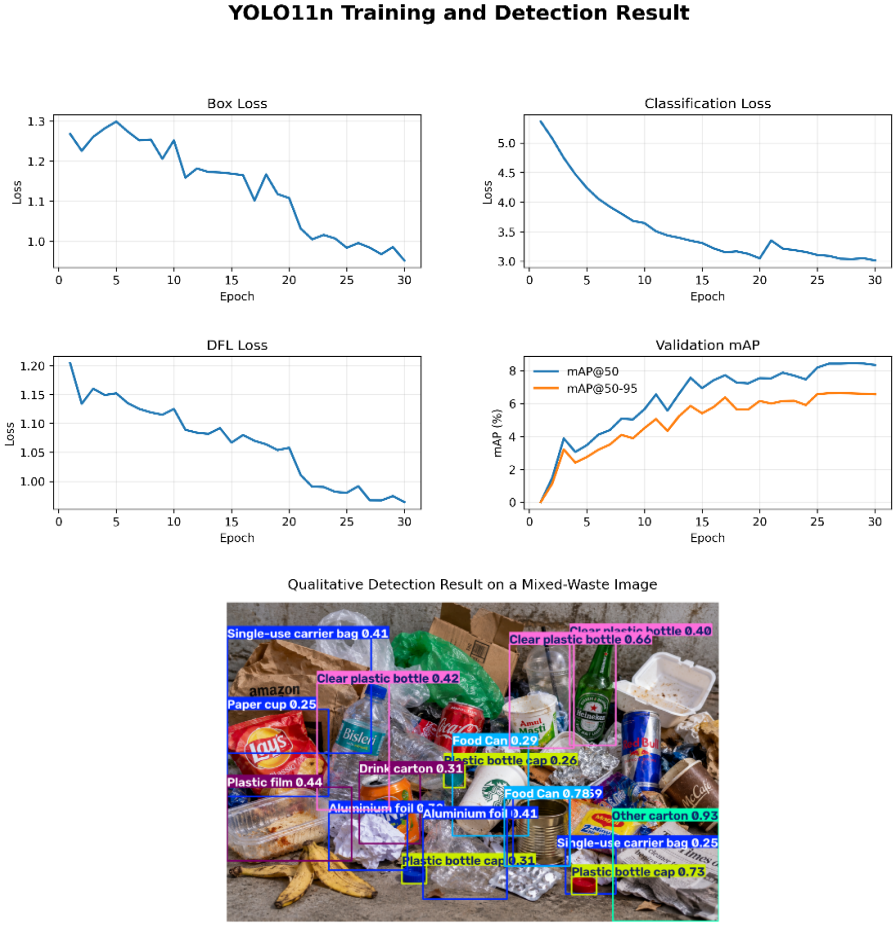


Figure 2: Yolo11n Training Set

The trained YOLO11n model successfully learned to detect multiple waste objects from complex images containing different object types, sizes, and backgrounds. During the initial stages of training, the detection performance was low, but the model showed gradual improvement in object localization and classification as training progressed. After 30 epochs, the best-performing checkpoint was selected based on validation performance.

For the final evaluation, the trained model was tested on the separate test subset containing 150 images. The model achieved a **Precision of 45.63%** and a **Recall of 11.83%**. The resulting **mAP@50 was 13.45%**, while the stricter **mAP@50-95 reached 11.44%**. These results indicate that the model was able to recognize several categories of waste objects, but its overall recall remained limited because of the large number of classes and the highly imbalanced distribution of annotations across them.

The class-wise results further demonstrated significant variation between waste categories. Categories with comparatively larger numbers of training examples, such as Clear plastic bottle, Drink can, Corrugated carton, and Aluminium foil, showed stronger detection performance than categories with extremely limited samples. For example, Clear plastic bottle achieved an mAP@50 of 35.9%, Drink can achieved 41.0%, and Corrugated carton achieved 50.7% on the validation set. In contrast, several rare categories produced very low or zero detection scores because of insufficient annotated examples.

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Table 3: Final Performance of YOLO11n on the Test Dataset

Metric	Result
Precision	45.63%
Recall	11.83%
mAP@50	13.45%
mAP@50-95	11.44%

The experimental results show that YOLO11n provides a functional baseline for multi-class waste object detection using the TACO dataset. The model was also capable of performing multi-object inference on unseen images, generating bounding boxes, predicted waste categories, confidence scores, and object counts. However, the relatively low recall and mAP values indicate that a considerable number of waste objects were not detected correctly, particularly for categories containing very few training examples.

Table 4: Representative Class-wise Validation Performance

Waste Category	Validation Instances	mAP@50
Clear plastic bottle	58	35.9%
Drink can	58	41.0%
Corrugated carton	16	50.7%
Aluminium foil	11	15.9%
Plastic bottle cap	54	14.9%
Plastic film	101	19.4%
Plastic straw	28	17.5%
Styrofoam piece	17	12.8%
Unlabeled litter	173	3.01%
Cigarette	114	1.15%

The class-wise analysis highlights the effect of class imbalance on detection performance. Although categories such as Cigarette and Unlabeled litter contain many annotations, their visual diversity and object characteristics still resulted in low mAP values. On the other hand, some categories with fewer samples achieved comparatively better scores, demonstrating that annotation quantity alone does not completely determine model performance.

The qualitative detection results demonstrate that the trained YOLO11n model can identify multiple waste objects simultaneously within a single image. The model generates bounding boxes around the detected objects and assigns each object a corresponding class label and confidence score. This enables the system to distinguish between different waste categories present in complex and mixed environments rather than assigning a single label to the entire image.

In the sample detection results, the model successfully identified several types of waste objects, including clear plastic bottles, food cans, plastic bottle caps, paper cups, plastic film, aluminium foil, cartons, and single-use carrier bags. Multiple instances of the same category could also be detected within a single scene. The bounding boxes provide the spatial location of each detected object, while the confidence values indicate the model's estimated certainty for the corresponding prediction.

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Figure 3: Sample Mixed-Waste Detection Result Using YOLO11n

These qualitative results demonstrate the practical operation of the proposed waste detection pipeline in a multi-object environment. Although some predictions were produced with relatively lower confidence and certain objects were missed or classified imperfectly, the model was able to detect and localize a variety of waste objects within cluttered scenes. Such detection capability can serve as a foundation for automated waste identification, object counting, and intelligent waste segregation applications, while the observed limitations indicate the need for further improvements in dataset balance, training strategy, and model optimization.

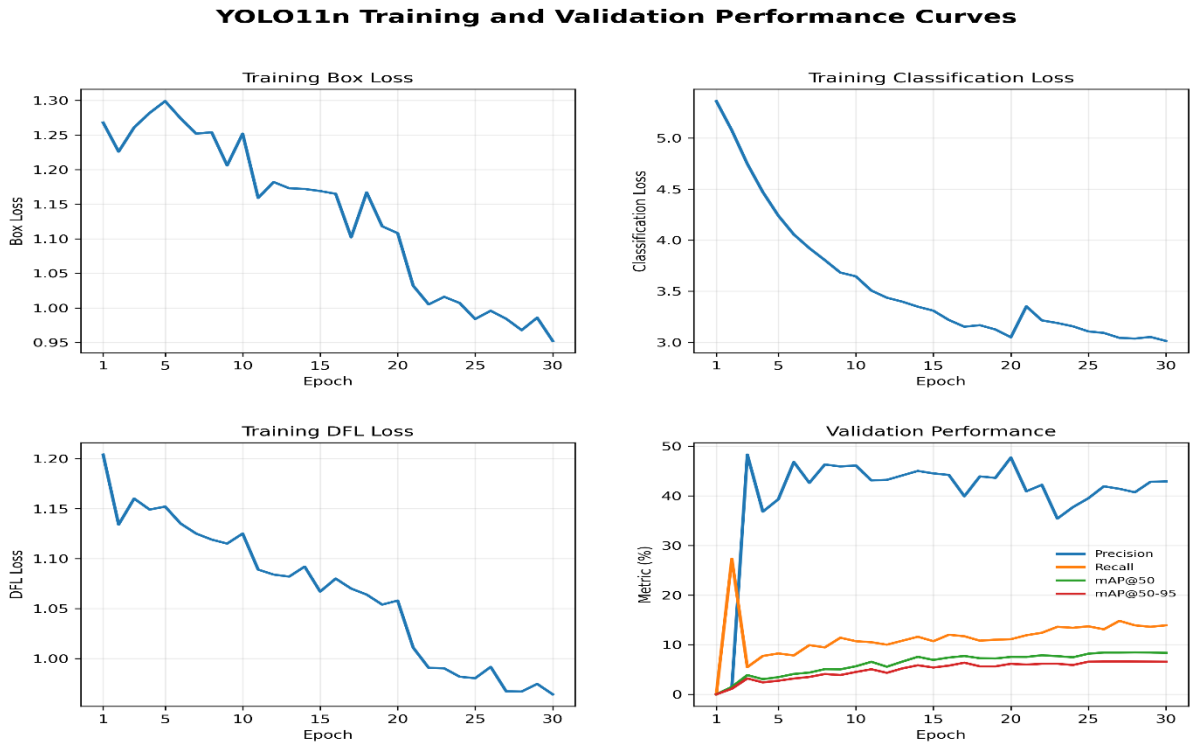


Figure 4: Training and Validation Performance Curves

The training process showed a reduction in box loss, classification loss, and distribution focal loss across the training epochs, indicating that the model progressively learned object localization and class-related features.

However, the validation metrics remained relatively low, reflecting the difficulty of learning 60 highly imbalanced waste categories from a comparatively small dataset.

VI. CONCLUSION

This study presented a deep learning-based approach for multi-class waste object detection using the YOLO11n architecture and the TACO dataset. A dataset containing 1,500 images, 4,784 annotated objects, and 60 waste categories was converted from COCO format into YOLO-compatible annotations and divided into training, validation, and testing subsets. The YOLO11n model was fine-tuned for 30 epochs using an NVIDIA Tesla T4 GPU and evaluated using Precision, Recall, mAP@50, and mAP@50-95.

The final evaluation on the test dataset produced a Precision of 45.63%, Recall of 11.83%, mAP@50 of 13.45%, and mAP@50-95 of 11.44%. The qualitative results demonstrated that the model can detect multiple waste objects within a single image and provide their corresponding locations, class labels, confidence values, and object counts. However, the results also highlighted the difficulty of detecting a large number of waste categories when the available annotations are highly imbalanced and several categories contain only a very small number of examples.

Overall, the findings establish YOLO11n as a baseline deep learning approach for multi-class waste object detection using real-world litter images. The current results provide a foundation for future improvements, including addressing class imbalance, increasing the number of annotated samples for rare waste categories, applying advanced data augmentation strategies, optimizing training parameters, and exploring improved object detection architectures. Such advancements can strengthen the applicability of deep learning-based systems for automated waste segregation, recycling assistance, and intelligent waste management applications.

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